SPE-229949-MS

Methodology for Quantifying Fluid Partition Coefficient Under Annular Self-Flow in Esp-Equipped Wells: Experimental and Numerical and Field Validation Insights

Evgeny V. Yudin, Rinat A. Khabibullin, Oleg S. Kobzar, Andrey V. Ryzhikov, Vladislav E. Chernyshov, Dmitry V. Usikov, Makhsud A. Suleymanov, Ivan V. Grigorev, Maxim D. Shabunin, and Mikhail V. Verbitsky, Independent Consultant

Copyright 2025, Society of Petroleum Engineers DOI 10.2118/229949-MS

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This work develops a methodology to quantify the liquid partition coefficient in wells equipped with electric submersible pumps (ESPs) operating under conditions of self-flow in the annular space. The liquid partition coefficient, determined here as a measure of how the total liquid flow divides between tubing and annulus, becomes particularly informative in high–gas-content environments where ESP performance can degrade and natural gas separation intensifies. Under such circumstances, redistribution of flow between the tubing and the annulus may occur, giving rise to operational anomalies that lie outside the conventional ESP operating envelope.

To address this problem, the study employs a comprehensive, multi-source approach that integrates controlled laboratory experiments, physics-based numerical modeling, and analysis of field measurements from a representative oil well. The laboratory component uses a multiphase air–water flow loop to reproduce key features of gas–liquid transport relevant to ESP operation and self-flow through the annulus. These experiments reveal a clear trend: the liquid partition coefficient decreases as the gas volume fraction increases, consistent with the strengthening of downhole gas separation processes and the resulting shift in how phases distribute between the two flow paths.

Building on these observations, a numerical algorithm is formulated to iteratively resolve the pressure distribution in both tubing and annulus over a range of boundary conditions. The scheme accounts for ESP performance degradation in the presence of free gas, for gas–liquid separation, and for the ensuing redistribution of phase flows. Validation against independent telemetry and multiphase flowmeter data demonstrates that the algorithm reproduces observed behavior with sufficient accuracy and reliability for practical use.

In application, the methodology enables quantitative assessment of phase-flow partitioning and supports early detection of self-flow through the annulus regimes that are typically excluded from standard ESP operating strategies. When integrated into monitoring and control systems, it offers a

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

pathway to real-time diagnostics and optimization of artificial-lift performance. The findings thus carry direct practical implications for well design, equipment selection, and mitigation of abnormal flow conditions in mechanized oil production.

Introduction

Modern conditions of oil and gas field development require the use of adaptive production technologies that take into account the peculiarities of deposits with high gas saturation, active gas caps and unconventional oil reserves. Gas and gas condensate fields with oil rims - thin layers of oil up to 30 meters thick located between the water-bearing and gas-bearing parts of the deposit - are a special category.

The operation of such fields involves a number of technical difficulties, particularly elevated gas content at the intake of submersible equipment. Incorrect selection of submersible equipment may lead to well self-flowing through the annulus. The occurrence of such cases requires detailed studies aimed at developing methods for optimizing production processes under these conditions.

Self-flowing through the annular space (FTA) can be observed in wells where the installed pumps have insufficient capacity. In such cases the pump operates in the right-hand region of the pressure-flow characteristic (PFC). According to the complete pressure-flow characteristic given in Stepanov [1], ESP in some cases may generate a negative pressure differential, act as a hydraulic restriction.

Research carried out laboratory bench [2] has demonstrated that similar patterns persist during the operation of submersible electric centrifugal pumps in gas–liquid flow conditions.

The article [3] presents the results of a laboratory study of electric submersible pump operation with gas-liquid mixture in the high-flow mode. The authors analyzed the operation of oil wells and developed models using the UniflocVBA software package describing the processes of self-flow through the annular space when the pump operates at high flow rates and negative pressure heads. As a result, the study revealed that at a high gas-oil ratio the pump is able to form negative pressure differential, which leads to redistribution of flows in the well. The authors developed a mathematical model that takes into account degradation of ESP performance characteristics with increasing gas volume fraction and defines the conditions under which self-flowing through the annular space is possible.

It should be taken into account that in addition to gas separation at the intake of the submersible equipment, fluid separation occurs between the flows in the tubing and annular space.

The liquid partition coefficient (K_{liq}) in this case can be determined as follows:

$$K_{liq} = \frac{Q_{liq}^{ann}}{Q_{liq}},$$

where:

- Q_{liq}^{ann} is the volume of fluid entering the annular space;
- Q_{lid} is the volume of fluid entering the well from the reservoir.

The fluid partition coefficient is defined in such a way that at its value equal to zero, all fluid flows into the tubing, as in the classical mode of well operation. When $K_{liq} = 1$, all fluid flows through the annular space [4].

Methodology

The authors conducted a study to investigate the dependence of the liquid partition coefficient on the flowing gas content using a "water-air" model mixture. The studies were carried out on a specialized stand simulating

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

the movement of a gas-liquid mixture through the tubing, production casing and wellhead equipment (see Fig. 1).

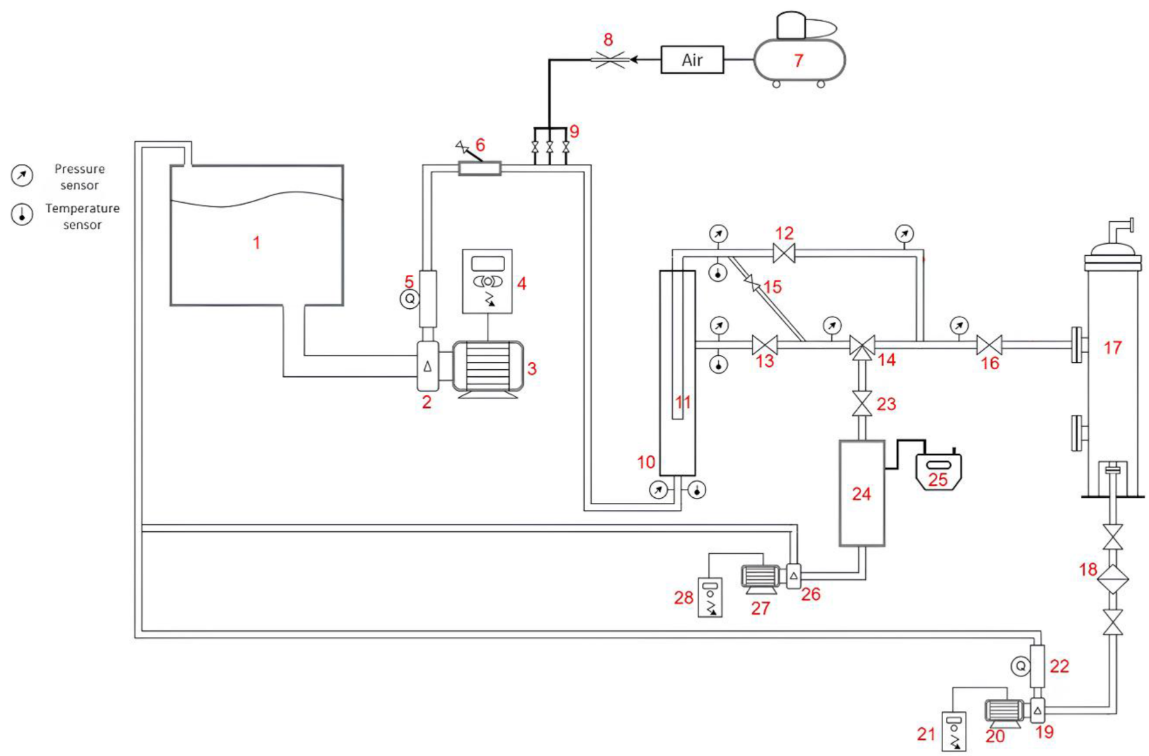

Figure 1—Schematic diagram of the laboratory bench.

Experiments were conducted in the range of liquid flow rates 75-150 m3/day. For each liquid flow rate, a certain gas flow rate was set so that the condition of maintaining volumetric gas volume fraction in the flow (30-90% vol.) was met. For all ranges of gas-liquid mixture the liquid and gas flow rates were measured using measuring devices shown in Fig. 1 (1 - tank; 2, 19, 26 - centrifugal pump; 3, 20, 27 - electric motor; 4, 21, 28 - controlled variable frequency drive; 5, 22 - mass flow meter; 6 – solid particle injection unit; 7 - piston compressor; 8 – gas rheometric flow meter; 9 - gas injection tubes; 10 - production casing; 11 - pump housing; 12 - tubing gate valve; 13 - pipe gate valve; 14 - three-way valve; 15 - bypass line with valve; 16 - line gate valve; 17 - cyclone separator; 18 - mechanical impurity filter; 23 - gate valve to separator; 24 - gravity separator; 25 - gas meter).

During the experiments, liquid was drawn from tank (1) to the intake of the centrifugal pump (2). Gate valves (12) and (13) were adjusted to maintain a constant pressure differential between the tubing and the annulus, thereby ensuring simultaneous flow of the gas–liquid mixture through both wellhead lines. The pump shaft speed was varied using the frequency converter (4) to establish the required liquid-delivery regimes, while gas was introduced by gradually increasing the gas-line pressure with the ball valves open on the rheometric bench, which increased the hydraulic resistance. The gas fraction in the stream was raised in tandem with fine adjustments of the main pump frequency and the position of gate valve (16). This protocol provided consistent, reproducible conditions for examining the dependence of the liquid partition coefficient on the gas volume fraction.

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

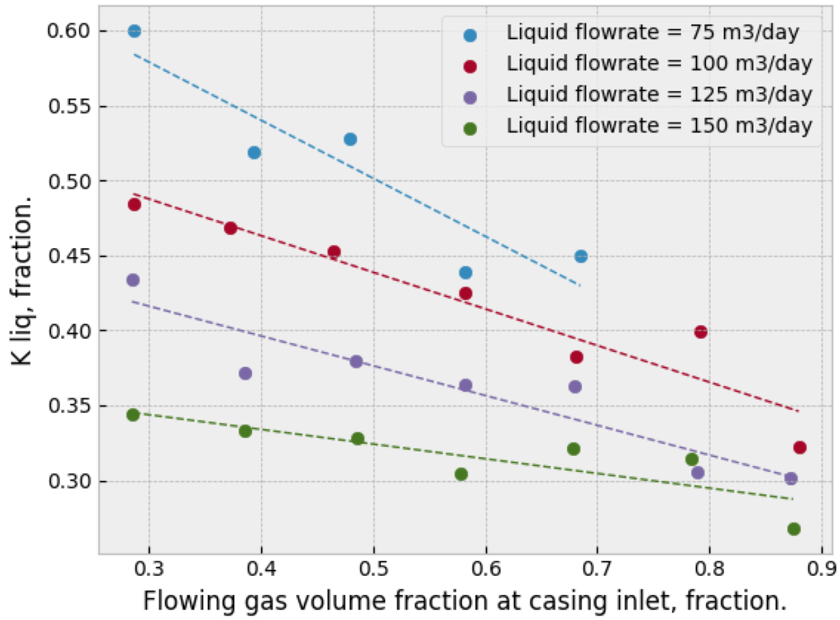

Figure 2—Graph of the liquid split ratio versus gas volume fraction at casing inlet.

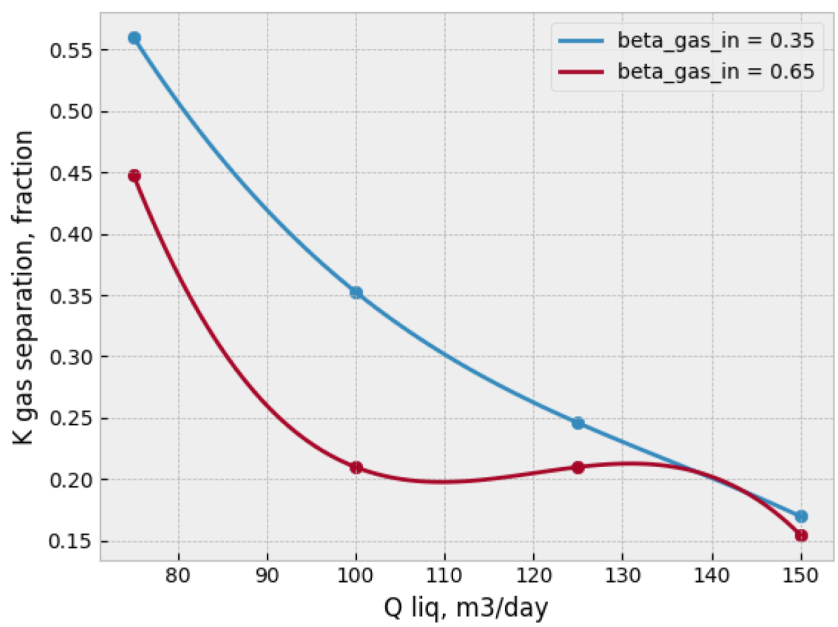

Figure 3—Graph of the natural gas separation coefficient at a fixed gas volume fraction.

Post-processing of the laboratory data reveals a consistent inverse relation between the liquid partition coefficient (K_{liq}) and the gas volume fraction (GVF) at the column inlet: as GVF increases, K_{liq} systematically declines. This downward trend is attributable to the effect of gas separation, which alters phase distribution and reduces the share of liquid directed along the annular path. The strength of this separation is, in turn, governed by the hydraulic regime: higher liquid flow rates shorten residence time and intensify turbulence, thereby weakening separation efficiency. Consequently, operating conditions characterized by a higher gas-separation factor (i.e., more effective natural separation) are associated with larger values of K_{liq}, whereas regimes with diminished separation reinforce the observed decrease of K_{liq} with increasing GVF. Together, these results indicate that K_{liq} is controlled by the interplay

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

between GVF and separation efficiency, with increased GVF driving K_{liq} downward and enhanced separation tending to raise it.

Modeling the operation of a well flowing through the annular space

A well exhibiting self-flow through the annulus can be represented by a coupled system of one-dimensional differential equations, partitioned into three domains that are linked by boundary conditions at their interfaces: from the bottomhole to the intake of the submersible equipment, the internal space of the tubing, and the annulus above the intake. In this regime the gas–liquid mixture is conveyed through two hydraulically separated channels – tubing and annulus – within which the PVT properties may differ because phase separation occurs at the pump intake. Consequently, the phase compositions of the streams are not identical, and the watercut in the tubing may differ from that in the annulus.

The modeling workflow proceeds by constructing pressure-distribution curves along each interval and updating them stepwise to reflect changes in key quantities: PVT properties, the oil- and water-partition coefficients together with the gas-separation factor, and the degradation of pump characteristics in the presence of free gas. Based on the block diagram presented in [4], a custom algorithm for modeling the FTA process was developed. It sequentially calculates the pressure distribution across three sections under shared boundary conditions and evaluates the phase-partition coefficients at the pump intake.

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

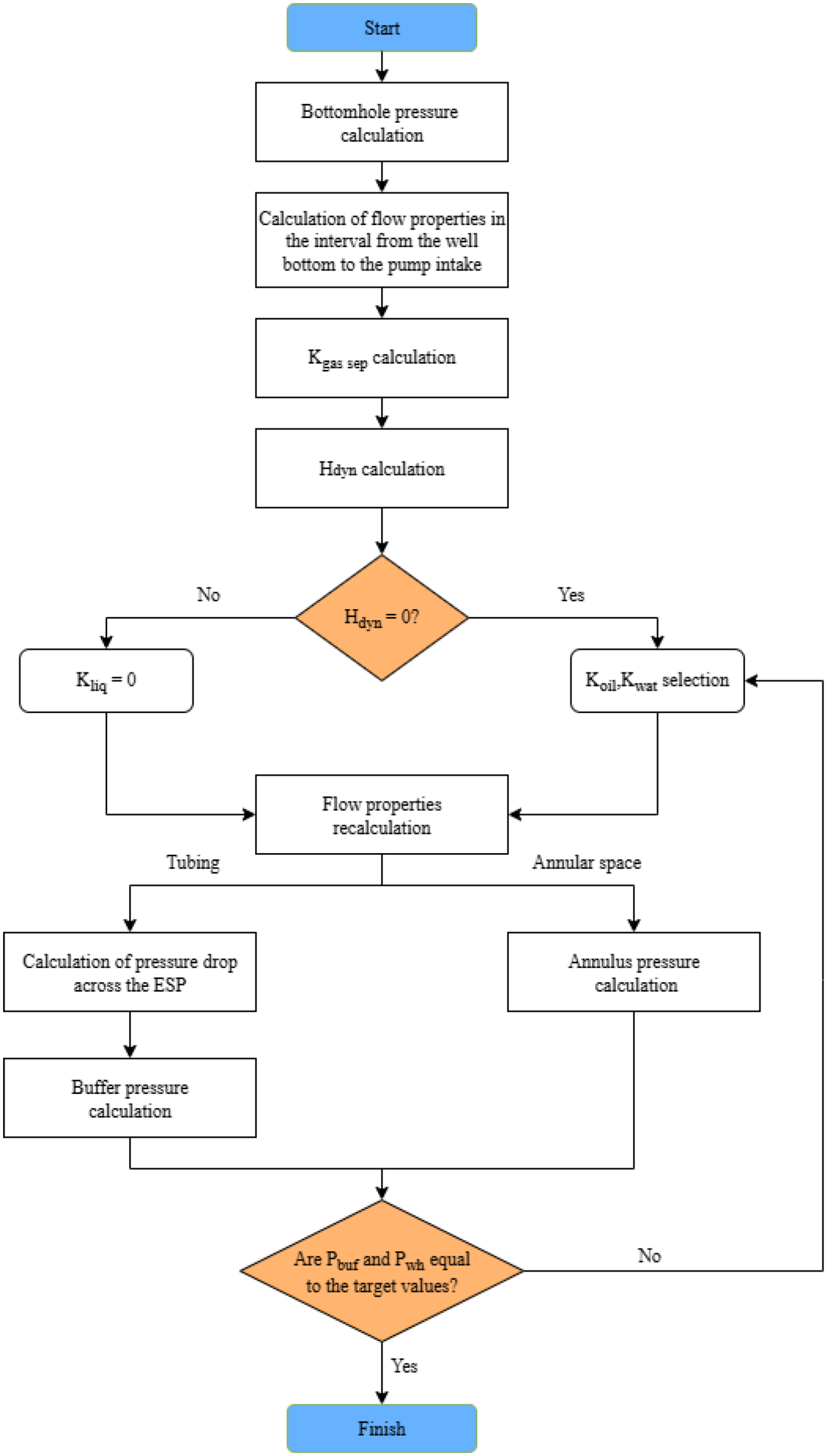

Figure 4—Calculation algorithm, where Kg.sep – gas separation factor, fr.; Hdyn – dynamic level, m; Kliq – liquid partition coefficient, fr.; Kwat – water partition coefficient, fraction.; Koil – oil partition coefficient, fr.; Pbuf – buffer pressure, atm; Pwh – annular pressure, atm.

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

Algorithm for calculating the liquid partition coefficient

The result of multiple calculations is a set of buffer and annular pressures for each oil splitting ratio, water splitting ratio and gas separation ratio, with fixed flow properties. A graph is plotted against the data obtained.

Using linear interpolation, a solution is found that satisfies the boundary conditions for annular and buffer pressures:

$$P_{buf}^{calc} = P_{buf}^{fact} \text{ and } P_{ann}^{calc} = P_{ann}^{fact}$$

The solution curves for tubing and annular space are plotted, where the intersection point corresponds to the desired solution (Fig. 5).

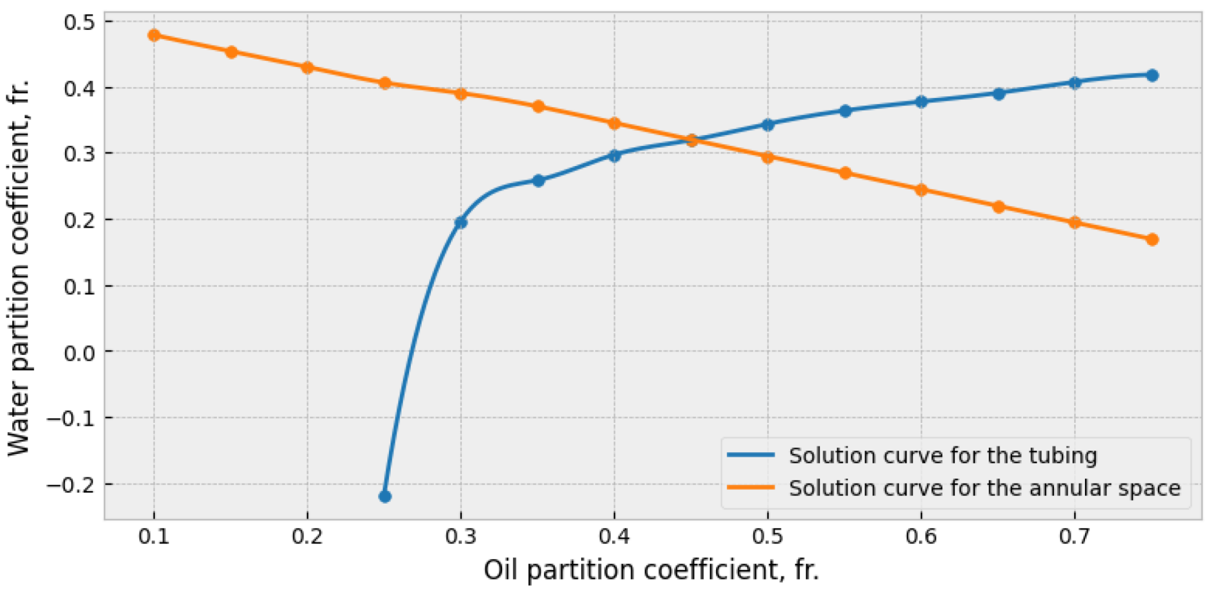

Figure 5—Solution curve for the tubing and annular space.

For approbation of the algorithm was chosen well of one of the fields of Western Siberia, which periodically observed self flowing through the annular space with installed ESP5-160-2150 and gas separator MNGSL5. Parameters of the investigated well are presented in Table 1.

Table 1—Well parameters

Casing diameter, mm	Tubing diameter, mm	Liquid flowrate, m^3/day	GOR, m^3/m^3	Well depth, m	Watercut,%
168	73	139	799	2100	65

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

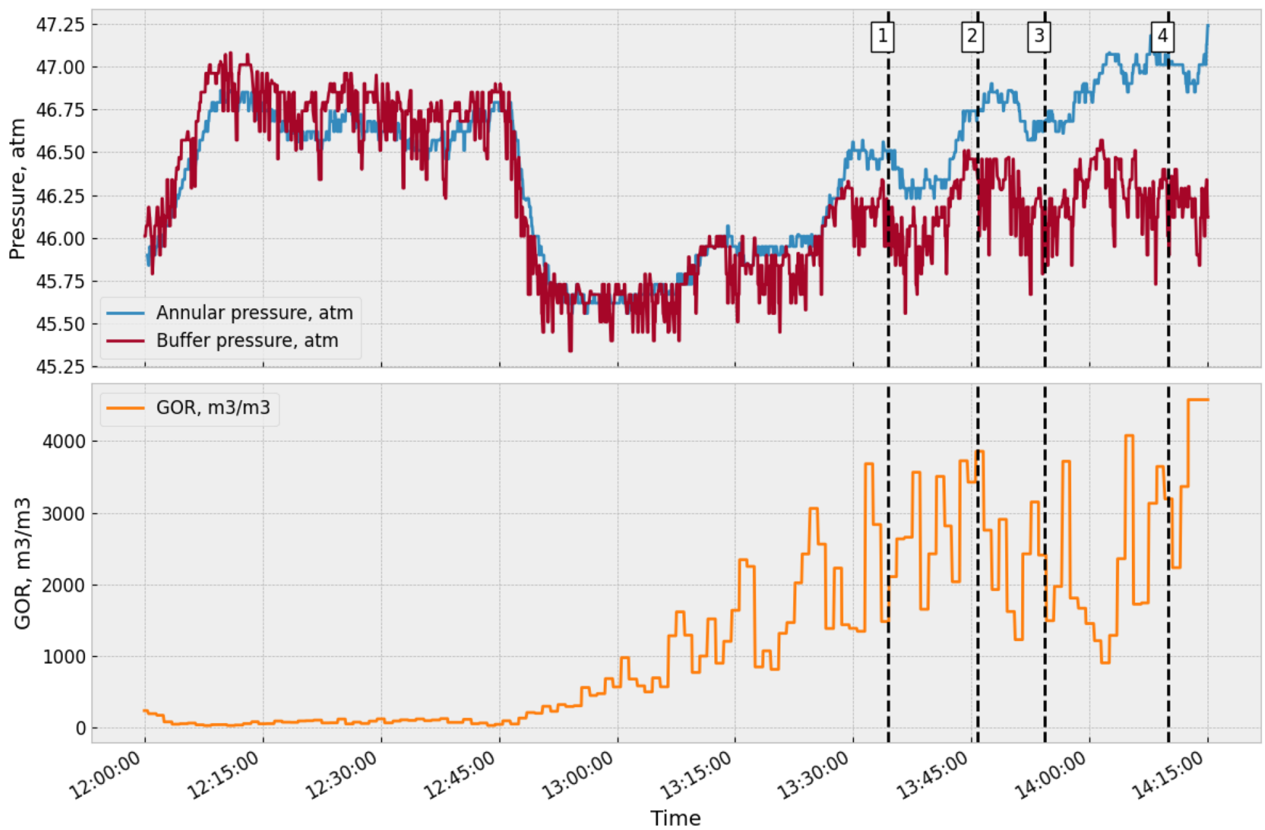

Figure 6—Field data analysis.

At 13:30 the difference between the annular and buffer pressures starts to appear, the subsequent increase in the pressure difference indicates the possible start of the FTA mode, as fluid starts to pass through the check valve of the christmas tree.

During normal ESP operation a dynamic level is established in the well, at that gas discharge from the annular space into the line occurs through the check valve on the surface, and the graph of annular pressure ($P_{ann}(t)$) practically coincides with the graph of buffer pressure ($P_{buf}(t)$), which is observed at the time interval 10:00-13:30 An increase in the differential pressure indicates the onset of liquid flow through the check valve.

According to the known values of the well operation parameters, the UniflocVBA software package was used to plot the pressure distribution curve (see Fig. 7), which is characteristic of the well operation mode when flowing through the annulus according to the algorithm described above. Blue color indicates the pressure distribution at the bottomhole-pump intake, calculated from bottom to top. Red - pressure drop across the pump. The violet line shows the bottom-up pressure distribution in the tubing from the pump outlet to the buffer. The green line indicates the bottom-up pressure distribution in the annular space from the pump intake to the wellhead interval.

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

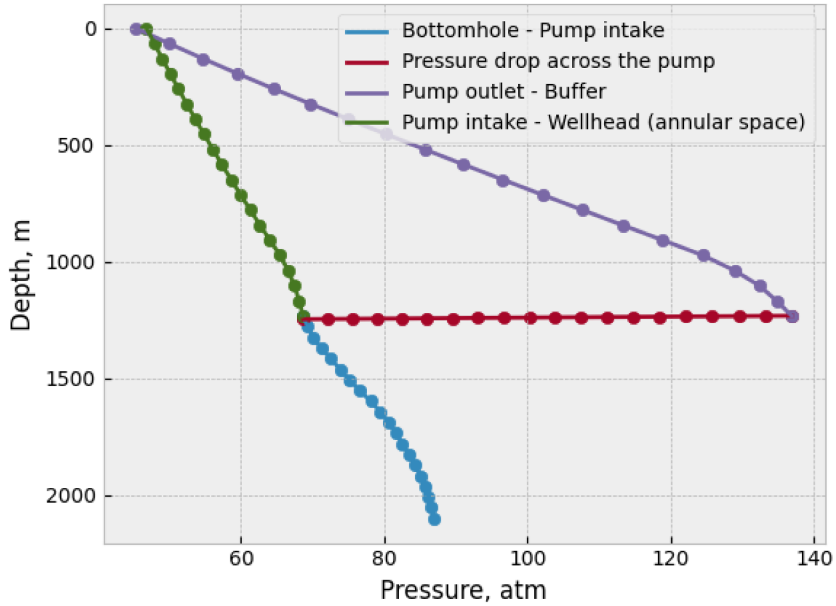

Figure 7—Pressure distribution curve in Well No. 1 during flow through the annular space.

The pressure distribution on the bottomhole - pump intake interval was calculated using the Ansari correlation [5]. The pressure drop in the columns after fluid separation was determined using the Gray correlation [6]. The gas separation factor was estimated based on the Marquez correlation [7]. Pump degradation was accounted for using an empirical model developed through bench testing [2].

Using the described algorithm, calculations were performed for different measurement points (see Fig. 6).

Table 2—Calculation for 4 points

№ point	1	2	3	4
Time	13:34:23	13:45:48	13:54:17	14:09:54
Q_{liq}, m³/day	189,8	209,8	151,63	133,12
Wc, %	93,33	97,23	92,72	93,1
Rp, m³/m³	1482	3425	2409	3643
P_{buf} atm	46,18	46,34	46,18	46,01
P_{ann}, atm	46,46	46,74	46,68	47,07
ΔP, atm	0,28	0,4	0,5	1,06
$P_{in.}$, atm	69,08	69,31	69,94	70,38
K_{oil},fraction	0,23	0,37	0,63	0,29
K_{wat}, fraction	0,11	0,12	0,41	0,43
K_{liq}, fraction	0,119	0,129	0,267	0,43

According to the obtained data, a graph is plotted (see Fig. 8), which shows a tendency to increase the fluid partition coefficient with increasing difference between annular and buffer pressures, indicating that the amount of fluid flowing through the annular space between tubing and casing is increasing.

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

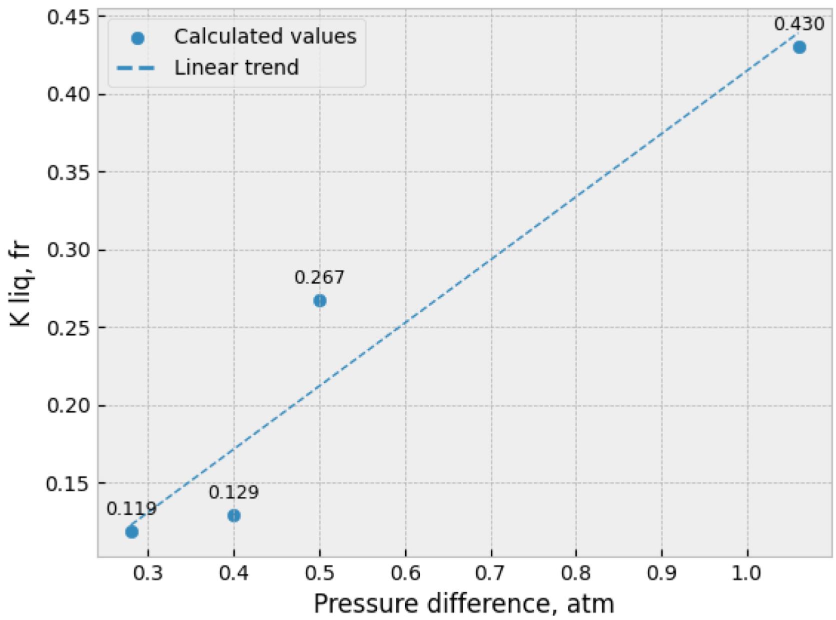

Figure 8—Variation of the liquid split ratio.

Results

The present work advances understanding of self-flow process in annular space in wells operated with ESPs. Its principal strength lies in an integrated design that combines controlled laboratory investigation, physics-based numerical modeling, and interpretation of measurements acquired from an oil field asset.

Using a water–air analog system, we established – for the first time – the dependence of the liquid partition coefficient (K_{liq}) on flowing gas content. The laboratory program showed that:

- K_{liq} declines as gas volume fraction increases.
- Gas separation governs the redistribution of phase fluxes between the tubing and the annulus.

These outcomes sharpen current views of downhole gas–liquid separation and inform revisions to existing self-flow through the annulus models.

An iterative calculation routine was also developed that ingests telemetry and multiphase-flowmeter readings to recover K_{liq}, thereby providing a quantitative split of mixture flow between annulus and tubing – an essential parameter for assessing well performance. Because conventional ESP operating practice treats self-flow through the annulus as an off-design condition, the algorithm can be used to evaluate equipment selection; when embedded in surveillance systems, it facilitates early recognition of transitions to this regime and timely adjustment of operating parameters to prevent abnormal behavior.

In parallel, a numerical simulation framework was formulated for the self-flow process through the annular space that comprehensively represents the key controls on wellbore hydraulics: contrasts in PVT properties between the tubing and the annulus; the influence of the liquid partition coefficient – including potential differences in watercut across the two flow paths – together with ESP performance degradation in the presence of free gas; and the boundary conditions that govern operating stability. Applying this framework improves the fidelity of flow predictions, supports defensible equipment choices and operating optimization, and ultimately enhances production efficiency while reducing the risk of abnormal events.

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

Discussion

The results of the study demonstrate that in wells with high gas content and degraded ESP performance, fluid redistribution into the annular space can occur. Laboratory experiments and modeling of field data confirm a clear dependence of the liquid partition coefficient on gas volume fraction. The observed decrease in this coefficient with increasing gas content indicates intensified gas separation, which must be considered when analyzing deviations in well performance.

The developed numerical algorithm enables rapid assessment of phase flow distribution and early detection of self-flow through the annulus conditions. Its implementation can improve equipment selection, optimize operating strategies, and reduce the risk of abnormal well behavior. Integration into digital monitoring and control systems – such as SCADA or digital twins – could support real-time diagnostics and predictive maintenance. Additionally, the model's adaptability allows for scenario analysis under varying production and reservoir conditions, offering valuable insights for field development planning. Further validation across a wider range of wells, fluids, and operating regimes will strengthen the robustness and generalizability of the proposed approach.

Conclusion

The analysis indicates that self-flowing through the annulus (FTA) arises from flow redistribution under elevated gas volume fraction, with the primary drivers being suboptimal operating characteristics of the selected ESP, intensified gas separation, and reduced pressure at the pump intake. Laboratory experiments consistently showed that the liquid partition coefficient (K_{liq}) decreases as GVF increases, reflecting a more pronounced separation effect. Numerical modeling of the FTA process confirmed the applicability of the developed iterative algorithm, which provides accurate estimates of K_{liq} and identifies the conditions under which FTA is likely to occur.

Field verification on an oil well corroborated the model's performance: pressure-monitoring data demonstrated the feasibility of timely prediction and prevention of FTA. Taken together, the quantified relationships and the proposed algorithm offer practical means to optimize ESP operation, avert abnormal flow regimes, and improve the overall efficiency of mechanized oil production.

Nomenclature

FTA – Self-flowing through the annulus.
ESP – Electric Submersible Pump.
PFC – Pressure-Flow Characteristic.
GVF – Gas volume fraction.
FC – Frequency Converter.
WC – Watercut.
GOR – Gas Oil Ratio.

References

1. Stepanoff A.J. Centrifugal and axial flow pumps //*Theory, design, and application*. – 1957.
2. Goridko K.A. (2023) Vliyanie izmenyayushchihsya svojstv gazozhidkostnoj smesi po dline nasosa na harakteristiki pogruzhnoj elektrocentrobezhnoj nasosnoj ustanovki. Doct. Diss. [*Effect of varying gas-liquid mixture properties along the pump length on the performance of a submersible electric centrifugal pumping unit. Doct. Diss.*]. Moscow. P. 246. (In Russ.)

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025

3. Goridko K., Kobzar O., Verbitsky V., Khabibullin R. Analysis of Self Flowing through Annulus of wells operated with Electric Submersible Pumps, Western and Eastern Siberia Fields Cases. *SPE Russian Petroleum Technology*. 2020.
4. Yudin, E.V., Gorbacheva, V.N., and N.A. Smirnov. Modeling and optimization of wells operating modes under annular flow conditions. *OIJ*. 2022; (**2022**): 122–126.
5. Ansari, A. M., Sylvester, N. D., Sarica, C., Shoham, O., & Brill, J. P. A Comprehensive Mechanistic Model for Upward Two-Phase Flow in Wellbores. *SPE Production & Facilities*. 1994; **9**(02): 143–151.
6. Gray, H.E., *Vertical Flow Correlation in Gas Wells: User's Manual for API 14B Surface Controlled Subsurface Safety Valve Sizing Computer Program*. 2nd Edition. M.: American Petroleum Institute, Dallas, TX; 1978.
7. Marquez R. *Modeling Downhole Natural Separation: Ph.D Diss*. Tulsa, The University of Tulsa, 2004.

Downloaded from https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP by Dmitry Kruglov on 09 November 2025